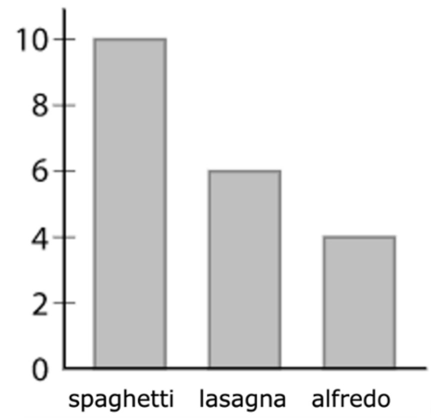
Grade 7 Name____________________
Unit 6 Date____________________
Lesson 10 Practice Period____________________

Su-an surveys a random sample of students about their favorite pasta dish served by the cafeteria and makes a bar graph of the results.
There are 1,826 students who attend the school. Use Su-an's data to answer the following questions.



1. Estimate the number of students at the school whose favorite dish is lasagna.

$\frac{6}{20} = \frac{3}{10}$ $\frac{3}{10} = \frac{x}{1826}$ $5478 = 10x$ $x = 547.8$ 548 students

2. Estimate the number of students at the school whose favorite dish is alfredo.

$\frac{4}{20} = \frac{1}{5}$ $\frac{1}{5} = \frac{x}{1826}$ $1826 = 5x$ $x = 365.2$ 365 students

3. Estimate the number of students at the school whose favorite dish is spaghetti.

$\frac{10}{20} = \frac{1}{2}$ $\frac{1}{2} = \frac{x}{1826}$ $1826 = 2x$ $x = 913$ 913 students

Jason is baking blueberry muffins with a recipe that calls for $2\frac{1}{2}$ cups (c) of flour and $\frac{3}{4}$ cup of sugar to make 9 large muffins. Jason wants to make as many large muffins as possible using 6 cups of flour.

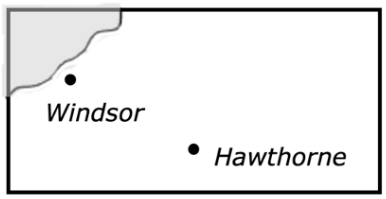
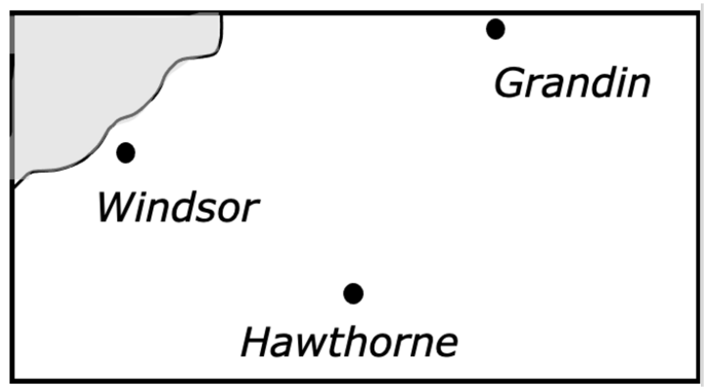
4. Exactly how many cups of sugar will Jason use if he uses all 6 cups of flour?

$\frac{2.5}{0.75} = \frac{6}{x}$ $4.5 = 2.5x$ $x = 1.8$ $1\frac{4}{5}$ cups

5. Exactly how many large muffins will Jason make if he uses all 6 cups of flour?

$\frac{2.5}{9} = \frac{6}{x}$ $54 = 2.5x$ $x = 21.6$ 21 muffins

The two maps show the same area at two different scales.

Map A	**Map B**
 1 cm = 2.75 mi	

Grandin is not on Map A, and Map B does not have a scale.
Windsor and Hawthorne are 1.8 cm apart on Map A.
Windsor and Hawthorne are 3.3 cm apart on Map B.
Windsor and Grandin are 5.6 cm apart on Map B.

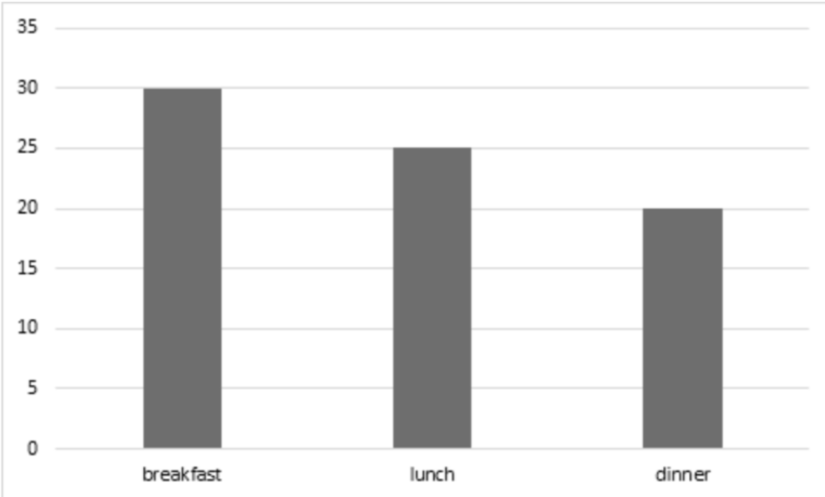
6. What is the distance, in miles, between Hawthorne and Windsor? Round to the nearest hundredth.

$$\frac{1\text{ cm}}{2.75\text{ miles}} = \frac{1.8\text{ cm}}{x\text{ miles}} \qquad x = 4.95 \text{ miles}$$

7. What is the distance, in miles, between Grandin and Windsor? Round to the nearest tenth.

$$\frac{3.3\text{ cm}}{4.95\text{ miles}} = \frac{5.6\text{ cm}}{x\text{ miles}} \qquad x = 8.4 \text{ miles}$$

Willow surveys a random sample of people in her town about their favorite food time, and makes a bar graph of the results, shown here:



If there are $2{,}400$ people living in her town, use the data from the graph to determine the following.

8. In total, there will be __800__ people in Willow's town whose favorite food time is lunch.

$\frac{25}{75} = \frac{1}{3}$ $\quad \frac{1}{3} = \frac{x}{2400}$ $\quad 2400 = 3x$ $\quad x = 800$

9. In total, there will be __640__ people in Willow's town whose favorite food time is dinner.

$\frac{20}{75} = \frac{4}{15}$ $\quad \frac{4}{15} = \frac{x}{2400}$ $\quad 9600 = 15x$ $\quad x = 640$

10. In total, there will be __960__ people in Willow's town whose favorite food time is breakfast.

$\frac{30}{75} = \frac{2}{5}$ $\quad \frac{2}{5} = \frac{x}{2400}$ $\quad 4800 = 5x$ $\quad x = 960$